

Renewable-Energy PoC - Technical Documentation

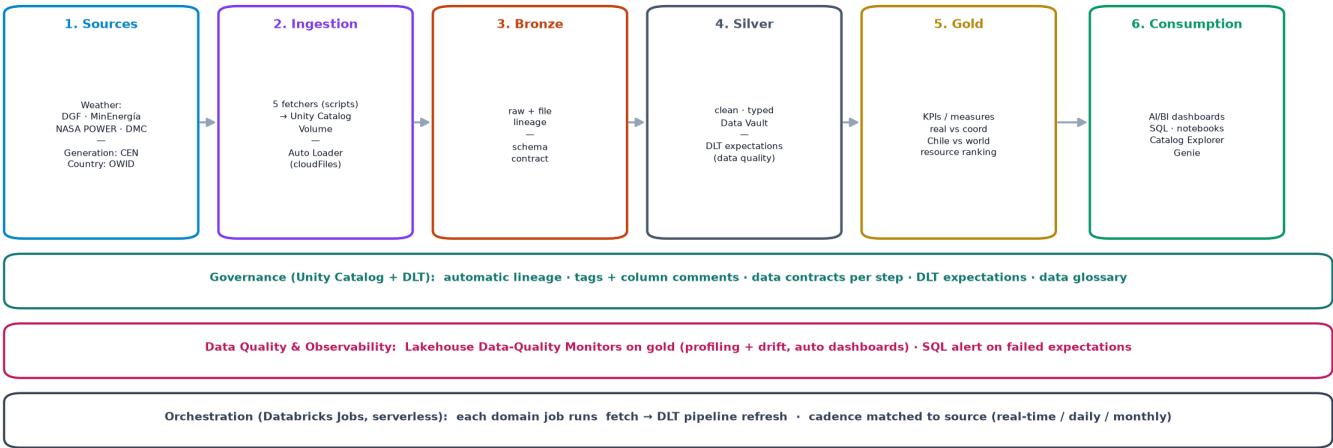
Medallion lakehouse on Databricks | updated 2026-07-21

1. Architecture

Databricks Asset Bundle with DLT on serverless, Auto Loader ingestion from UC Volumes, UC governance. Sources -> Bronze -> Silver -> Gold -> Consumption, orchestrated by Jobs and monitored with Lakehouse Data-Quality Monitors.

Renewable-Energy PoC — Data Architecture (Medallion on Databricks)

4 data sources · Unity Catalog Volume → Auto Loader → Bronze → Silver → Gold → Consumption · governed & monitored, orchestrated by Databricks Jobs

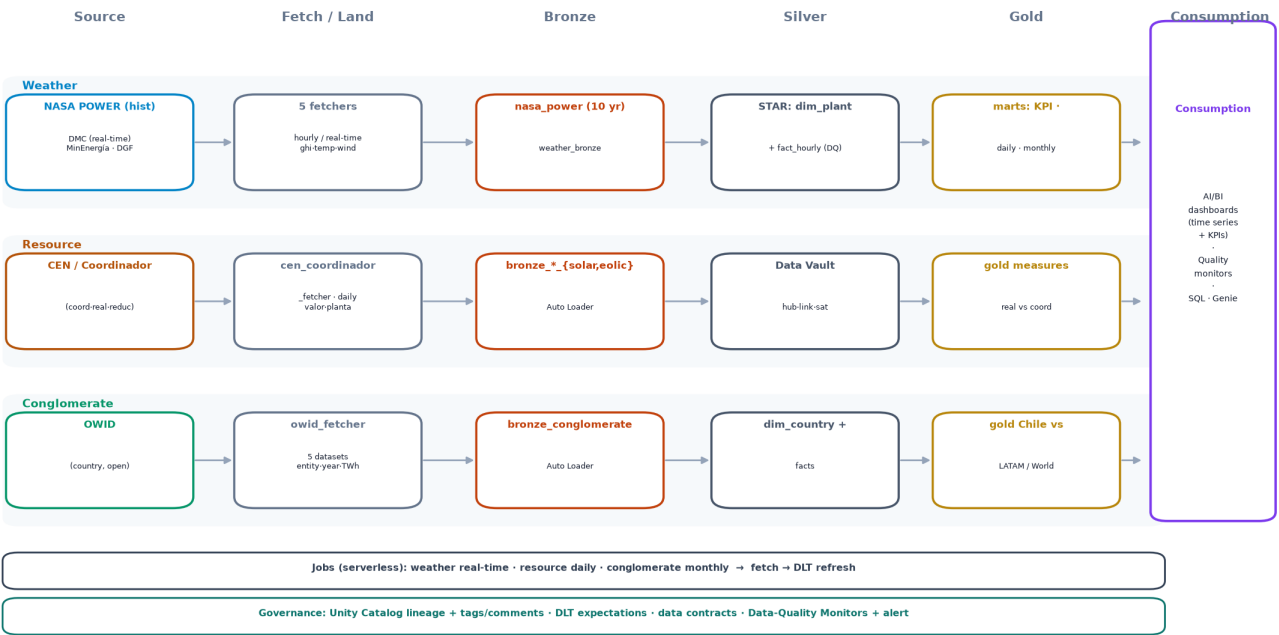


Databricks Asset Bundle (DAB) · three domains: Weather · Resource (generation) · Conglomerate (country) · everything in dev + branches

Weather sources by role: NASA POWER = historical hourly (10 yr) · DMC = real-time measured · MinEnergia/DGF = modeled resource

Renewable-Energy PoC — End-to-End Data Workflow

Per domain: source → fetcher → UC Volume → Bronze → Silver → Gold → Consumption · orchestrated by Databricks Jobs



2. Domains & sources

- Weather: NASA POWER (historical) + DMC (real-time) + MinEnergia/DGF (modeled).
- Resource: CEN generation (coord/real/reduc); silver = Data Vault.

- Conglomerate: OWID country stats.

3. Weather NASA - star schema + marts

- Fetcher: nasa_power_fetcher.py (hourly by lat/lon, yearly chunks, DMC-equivalent fields).
- Full load: 876,720 rows (10 plants x 10 years, 2015-2024).
- Silver STAR: silver_weather_nasa_dim_plant + silver_weather_nasa_fact_hourly (DLT expect_all_or_drop).
- Gold marts: resource_kpi (KPI/rank/tier) + daily (36,530) + monthly (1,200).
- AI/BI dashboard: monthly GHI/wind per plant + KPI table.
- Reproducible as the DLT pipeline weather_nasa (bundle validate OK).

4. Governance, DQ & observability

- Contracts per step + DLT expectations on silver + UC constraints.
- UC lineage + column comments + data glossary.
- Three Data-Quality Monitors on gold + SQL alert.

5. Deploy & run (dev only)

databricks bundle deploy -t dev --var=dev_catalog=workspace

All work is in dev and on branches. Prod is managed by the project lead.

Verify the medallion & jobs

- Resource pipeline: <https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/9bcfc32a-19f0-4327-b5bc-e8ee1cc4a3a4?o=7474645896934260>
- Conglomerate pipeline: <https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/b5d63282-a439-42ef-8e3b-cd707a9d5f58?o=7474645896934260>
- Weather NASA dashboard (series+KPIs): <https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18771593d19568e23e322277007c3?o=7474645896934260>
- Weather NASA gold monthly: https://dbc-54b27bae-2e91.cloud.databricks.com/explore/data/workspace/dev_fuad_onate_renewable_gold_energy_chile/gold_weather_nasa_monthly?o=7474645896934260
- Quality monitor (Resource gold): <https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f1809adb1a199384b80e96c469a206?o=7474645896934260>
- Pull requests: https://github.com/oxiboy/poc_databricks_evaluateserve/pulls